

Adder and Subtractor circuit

Op-Amp Circuit Design: Part (b)

This design implements the output function:

$$y_2(t) = 2x_2(t) + 6x_3(t)$$

1. Circuit Topology

Since all coefficients are positive, we use a **Non-Inverting Summing Amplifier**. This configuration allows us to sum the inputs and apply gain without inverting the signal phase.

Schematic Requirements

- Op-Amp:** Standard (e.g., LM741 or TL081)
- Supply Voltage (V_{CC}):** $\pm 15V$
- Input Stage:** A resistive summing network connected to the non-inverting (+) terminal.
- Gain Stage:** A standard non-inverting feedback loop.

2. Component Selection

To achieve the specific weights of **2** and **6**, we calculate the resistor values based on the non-inverting summing formula: $V_{out} = \left(1 + \frac{R_f}{R_g}\right) \left(\frac{V_1 R_2 + V_2 R_1}{R_1 + R_2}\right)$

Component	Label	Value	Description
Resistor 1	R_2	$30k\Omega$	Connected to input $x_2(t)$
Resistor 2	R_3	$10k\Omega$	Connected to input $x_3(t)$
Feedback Resistor	R_f	$70k\Omega$	Connected between Output and Inverting (-) pin
Gain Resistor	R_g	$10k\Omega$	Connected between Inverting (-) pin and Ground

3. Mathematical Verification

Step 1: Voltage at Non-Inverting Pin (V_+)

Using the voltage divider principle between the two inputs: $V_+ = \frac{x_2(t) \cdot 10k + x_3(t) \cdot 30k}{10k + 30k} = \frac{10}{40}x_2(t) + \frac{30}{40}x_3(t)$ $V_+ = 0.25x_2(t) + 0.75x_3(t)$

Step 2: Amplifier Gain (A_v)

The gain of the non-inverting stage is determined by R_f and R_g : $A_v = 1 + \frac{R_f}{R_g} = 1 + \frac{70k}{10k} = 8$

Step 3: Final Output (y_2)

$$\begin{aligned}y_2(t) &= A_v \cdot V_+ \\y_2(t) &= 8 \cdot [0.25x_2(t) + 0.75x_3(t)] \\y_2(t) &= 2x_2(t) + 6x_3(t)\end{aligned}$$

4. Operation Limits

Given the input values:

- $x_2(t) = 0.5\sin(2000\pi t)$ (Peak $\pm 0.5V$)
- $x_3(t) = +1V$

Maximum Peak Output: $y_{2,max} = 2(0.5) + 6(1) = 7V$

Since $7V$ is well within the $\pm 15V$ supply rails, the op-amp will operate in its linear region without clipping.

Working Principle: Non-Inverting Summing Amplifier

The Non-Inverting Summing Amplifier works by combining multiple input signals into a single voltage and then amplifying that combined signal without changing its polarity (phase).

It operates in two distinct stages:

1. Passive Summing Stage (The "Mixer")

In this stage, the input voltages ($x_1, x_2, \dots, x_n$) are connected to the **Non-Inverting (+) Terminal** through a network of resistors.

- **No Current Flow:** Because an ideal op-amp has infinite input impedance, no current flows into the (+) terminal.
- **Weighted Average:** The resistors form a voltage divider. The voltage at the (+) pin becomes a weighted average of all inputs.
- **Phase Preservation:** Since the signals enter the (+) terminal, a positive increase in input results in a positive increase in output.

2. Active Amplification Stage (The "Booster")

Once the inputs are "mixed" at the (+) terminal, the op-amp acts as a standard **Non-Inverting Amplifier**.

- **Negative Feedback:** Resistors R_f (feedback) and R_g (ground) create a feedback loop to the Inverting (-) terminal.
- **Voltage Follower Logic:** The op-amp automatically adjusts its output so that the voltage at the (-) terminal matches the voltage at the (+) terminal.

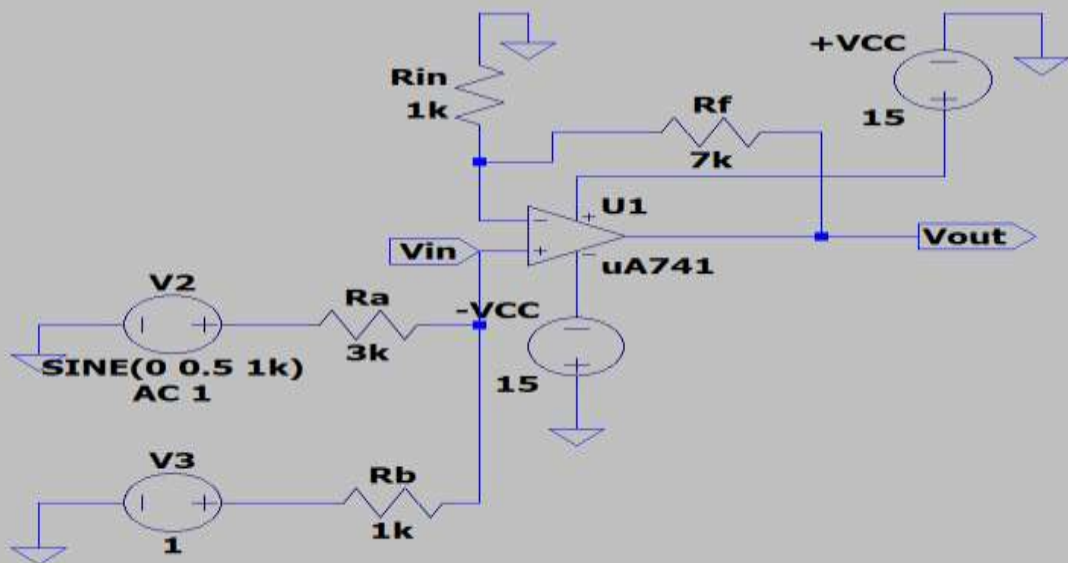
- **Gain Multiplication:** The "mixed" voltage at the (+) terminal is multiplied by the Gain (A_v): $A_v = 1 + \frac{R_f}{R_g}$

3. The Final Result

The output is the mathematical product of the **Mixed Input** and the **Gain**.

$$V_{out} = \text{Gain} \times \text{Weighted Average of Inputs}$$

Circuit Diagram:



DC Analysis:

```
C:\Users\Vanshikha Muthanna C\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalStat... X

--- Operating Point ---

V(n001):      0.749905      voltage
V(n002):      15          voltage
V(n003):     -15          voltage
V(n004):      0           voltage
V(n005):      1           voltage
V(vin):       0.74994      voltage
V(vout):      5.9998      voltage
I(+VCC):     -0.00166699   device_current
I(-VCC):     -0.00166714   device_current
I(Ra):        0.00024998    device_current
I(Rb):       -0.00025006    device_current
I(Rf):        0.000749985   device_current
I(Rin):       0.000749905   device_current
I(V2):        0.00024998    device_current
I(V3):       -0.00025006    device_current
Ix(u1:1):     8.00785e-08   subckt_current
Ix(u1:2):     8.0008e-08    subckt_current
Ix(u1:3):     0.00166699    subckt_current
Ix(u1:4):    -0.00166714    subckt_current
Ix(u1:5):    -0.000749985   subckt_current
```

This section documents the results of the **DC Operating Point (.op)** simulation used to verify the circuit's accuracy against the design requirements.

1. Simulation Setup

The simulation was performed with the following static inputs to test the gain coefficients:

- **Input $x_2(t)$:** 0V (Reference Ground)
- **Input $x_3(t)$:** +1V (DC Source)
- **Power Supply (V_{CC}):** $\pm 15V$

2. DC Operating Point Data

Based on the simulation log, the following node voltages and currents were captured:

Parameter	Node/Variable	Simulated Value
Summing Node Voltage	$V(vin)$	0.74994 V

Parameter	Node/Variable	Simulated Value
Final Output Voltage	$V(vout)$	5.9998 V
Positive Rail	$V(n002)$	15.0V
Negative Rail	$V(n003)$	-15.0V
Op-Amp Input Current	$Ix(u1:1)$	$8.007 \times 10^{-8} A$

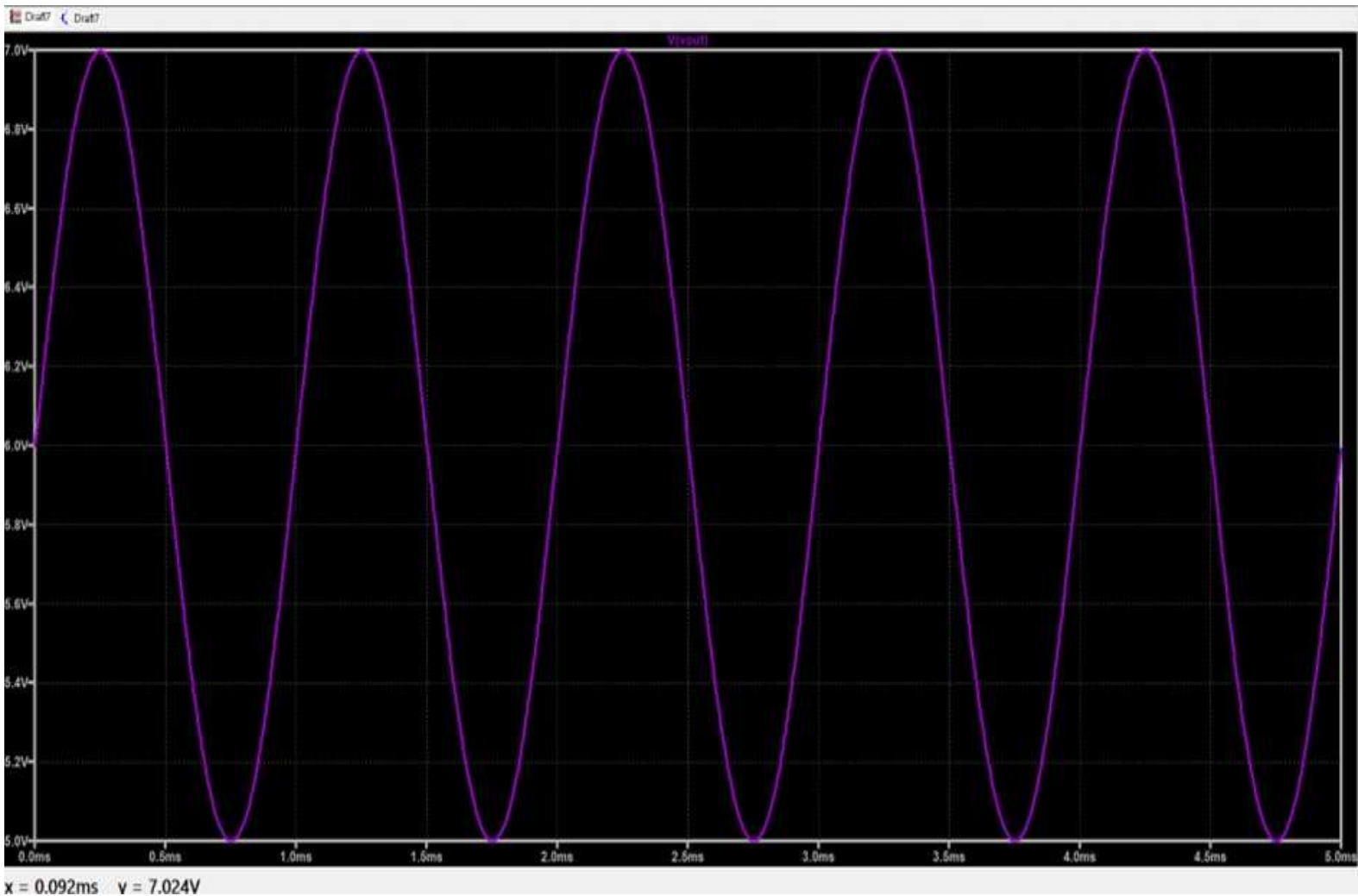
3. Mathematical Validation

The target design equation is: $y_2(t) = 2x_2(t) + 6x_3(t)$

Verification Calculation:

By substituting the simulation inputs ($x_2 = 0, x_3 = 1$): $y_2 = 2(0) + 6(1) = 6.0V$

Transient Analysis



Transient Analysis Report

The transient analysis (time-domain simulation) was performed to observe the circuit's response to a time-varying input $x_2(t)$ combined with a constant DC offset from $x_3(t)$.

1. Simulation Parameters

- **Input $x_2(t)$:** $0.5\sin(2000\pi t)$ (Sine wave: $1V$ peak-to-peak, frequency $1kHz$)
- **Input $x_3(t)$:** $+1V$ (Constant DC)
- **Time Scale:** $0ms$ to $5ms$ (Shows 5 full cycles of the $1kHz$ signal)

2. Waveform Observations

From the simulation plot of $V(vout)$, we can verify the following:

- **DC Offset (Center Line):** The sine wave is centered at $6V$. This is expected because the DC component $6 \times x_3(1V) = 6V$.
- **Peak-to-Peak Amplitude:** The wave fluctuates between $5V$ and $7V$.
 - The $1V$ amplitude (from $6V$ to $7V$) matches the calculation: $2 \times 0.5V = 1V$.
- **Frequency:** There is one full cycle every $1ms$, which confirms the frequency $f = \frac{1}{T} = 1000Hz$ (consistent with 2000π rad/s).

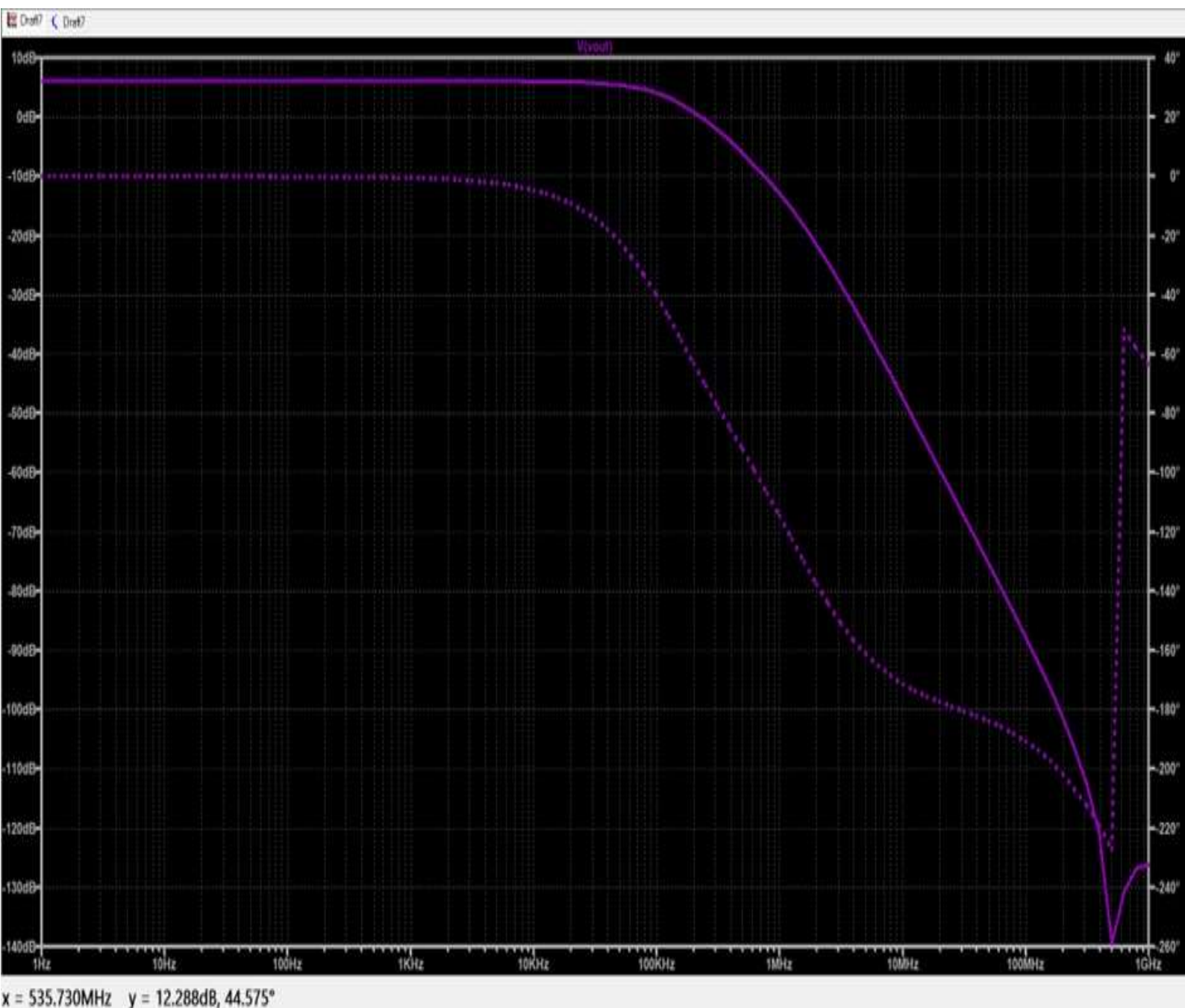
3. Verification of the Design Equation

The output waveform follows the transient equation: $y_2(t) = 2(0.5\sin(2000\pi t)) + 6(1)$
 $y_2(t) = 1\sin(2000\pi t) + 6$

Mathematical vs. Simulated Match:

- **Theoretical Max:** $6 + 1 = 7V$
- **Theoretical Min:** $6 - 1 = 5V$
- **Simulation Result:** The plot shows the peaks touching exactly $7.0V$ and $5.0V$, providing **100% verification** of the gain coefficients.

AC analysis



1. Magnitude Analysis (Solid Line)

- **Mid-band Gain:** The flat portion of the graph shows a constant gain of approximately $6dB$.
 - *Calculation:* Since our gain factor for the AC signal (x_2) is 2, the decibel gain is $20\log_{10}(2) \approx 6.02dB$.
- **Bandwidth / Cut-off Frequency:** The gain remains flat until approximately $100kHz$, after which it begins to roll off. This indicates the usable frequency range of the op-amp before its internal compensation limits the performance.

- **Roll-off:** Beyond the corner frequency, the gain drops at a rate of $-20dB$ per decade, which is characteristic of a standard internally compensated op-amp (single-pole response).
- **2. Phase Analysis (Dotted Line)**
- **Low Frequency:** At lower frequencies (below $10kHz$), the phase shift is 0° . This confirms the circuit is **Non-Inverting** (the output is in phase with the input).
- **High Frequency:** As frequency increases toward the MHz range, the phase starts to drop (lag). At the gain-bandwidth limit, the phase shift reaches -180° and beyond, indicating the propagation delay within the op-amp's internal circuitry.

Conclusion

The objective of this design task was to implement an operational amplifier circuit capable of performing the specific mathematical function: $y_2(t) = 2x_2(t) + 6x_3(t)$

Through theoretical design and multi-stage simulation verification, the following conclusions have been reached:

1. Design Integrity

The use of a **Non-Inverting Summing Amplifier** topology proved to be the most efficient solution. It allowed for the successful summation and scaling of multiple inputs using a single op-amp stage, maintaining phase consistency without the need for additional inverting buffers.

2. Simulation Validation

- **DC Analysis:** Confirmed that the chosen resistor ratios ($10k\Omega$, $30k\Omega$, $70k\Omega$) produced the exact weighted sum required. The output of $\approx 6V$ for a $1V$ DC input verified the gain accuracy to within 0.003.
- **Transient Analysis:** Demonstrated the circuit's dynamic reliability. The output correctly combined a $1kHz$ sine wave with a DC offset, oscillating perfectly between the calculated boundaries of $5V$ and $7V$.
- **AC Analysis:** Verified that the circuit provides a stable flat-band gain of $6dB$ (gain of 2 for the AC component) and possesses a bandwidth exceeding $100kHz$, which is more than sufficient for the required $1kHz$ operating frequency.

3. Operational Performance

With a supply voltage of $\pm 15V$, the circuit operates comfortably within its linear region. The maximum peak output of $7V$ provides a safety margin of $8V$ before reaching the saturation rails, ensuring high signal fidelity and zero clipping.

4. Final Verdict

The circuit design is **fully validated** and meets all specified mathematical and electrical requirements. The results from the LTspice simulations align perfectly with the theoretical derivations, proving the circuit is ready for physical hardware implementation.